

Terminal - osboxes@osboxe...

Terminal - osboxes@osboxes:~

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osboxes@osboxes:~$ ip a | grep inet
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host noprefixroute
    inet 10.0.2.4/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
    inet6 fe80::c042:6e93:bf05:e726/64 scope link noprefixroute
osboxes@osboxes:~$ echo "10.0.2.4 khadimul-islam.me" | sudo tee -a /etc/hosts
> sudo hostnamectl set-hostname khadimul-islam.me
>
```

```
Executing: /usr/lib/systemd/systemd-sysv-install enable mysql
osboxes@khadimul-islam:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-11-10 23:20:16 EST; 3min 28s ago
  Invocation: 612b5f15304e45ef84cde32262ec1e8c
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 10827 (apache2)
      Tasks: 6 (limit: 1065)
    Memory: 12.4M (peak: 12.6M)
       CPU: 190ms
    CGroup: /system.slice/apache2.service
            └─10827 /usr/sbin/apache2 -k start
              └─10831 /usr/sbin/apache2 -k start
                └─10832 /usr/sbin/apache2 -k start
                  └─10833 /usr/sbin/apache2 -k start
                    └─10834 /usr/sbin/apache2 -k start
                      └─10835 /usr/sbin/apache2 -k start

Nov 10 23:20:16 khadimul-islam.me systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 10 23:20:16 khadimul-islam.me systemd[1]: Started apache2.service - The Apache HTTP Server.
osboxes@khadimul-islam:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-11-10 23:19:59 EST; 4min 0s ago
  Invocation: 6bb69bc8aae344bbbd9fd10600658308
     Main PID: 9158 (mysqld)
    Status: "Server is operational"
      Tasks: 37 (limit: 1065)
    Memory: 176.7M (peak: 379.1M swap: 175.8M swap peak: 175.8M)
       CPU: 6.302s
    CGroup: /system.slice/mysql.service
            └─9158 /usr/sbin/mysqld

Nov 10 23:19:57 khadimul-islam.me systemd[1]: Starting mysql.service - MySQL Community Server...
Nov 10 23:19:59 khadimul-islam.me systemd[1]: Started mysql.service - MySQL Community Server.
osboxes@khadimul-islam:~$
```

```
osboxes@khadimul-islam:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.42-0ubuntu0.24.10.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| appdb    |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
5 rows in set (0.08 sec)

mysql> SELECT user, host FROM mysql.user;
+-----+-----+
| user          | host          |
+-----+-----+
| appuser       | localhost    |
| debian-sys-maint | localhost    |
| mysql.infoschema | localhost    |
| mysql.session | localhost    |
| mysql.sys     | localhost    |
| root          | localhost    |
+-----+-----+
6 rows in set (0.01 sec)

mysql> exit;
Bye
osboxes@khadimul-islam:~$
```

```
Terminal - osboxes@khadimul-islam: ~
File Edit View Terminal Tabs Help
osboxes@khadimul-islam:~$ getent passwd appManager || \
sudo adduser --system --home /nonexistent --no-create-home --shell /bin/bash appManager
[sudo] password for osboxes:
Info: The home dir /nonexistent you specified can't be accessed: No such file or directory

Info: Selecting UID from range 100 to 999 ...

osboxes@khadimul-islam:~$ echo 'appManager ALL=(root) NOPASSWD:/usr/bin/systemctl start apache2,/usr/bin/systemctl stop apache2,/usr/bin/systemctl restart apache2,/usr/bin/systemctl status apache2,/usr/bin/systemctl start mysql,/usr/bin/systemctl stop mysql,/usr/bin/systemctl restartmysql,/usr/bin/systemctl status mysql' | sudo tee /etc/sudoers.d/appManager
appManager ALL=(root) NOPASSWD:/usr/bin/systemctl start apache2,/usr/bin/systemctl stop apache2,/usr/bin/systemctl restart apache2,/usr/bin/systemctl status apache2,/usr/bin/systemctl start mysql,/usr/bin/systemctl stop mysql,/usr/bin/systemctl restartmysql,/usr/bin/systemctl status mysql
osboxes@khadimul-islam:~$ sudo chmod 440 /etc/sudoers.d/appManager
osboxes@khadimul-islam:~$
```

```
osboxes@khadimul-islam:~$ sudo sed -i 's/^export APACHE_RUN_USER=.*$/export APACHE_RUN_USER=webuser/' /etc/apache2/envvars
osboxes@khadimul-islam:~$ sudo sed -i 's/^export APACHE_RUN_GROUP=.*$/export APACHE_RUN_GROUP=webgroup/' /etc/apache2/envvars
osboxes@khadimul-islam:~$ sudo mkdir -p /var/www/khadimul-islam
osboxes@khadimul-islam:~$ echo "<?php phpinfo(); ?>" | sudo tee /var/www/khadimul-islam/index.php > /dev/null
osboxes@khadimul-islam:~$ sudo chown -R webuser:webgroup /var/www/khadimul-islam
osboxes@khadimul-islam:~$ sudo nano /etc/apache2/sites-available/khadimul-islam.me.conf
osboxes@khadimul-islam:~$ sudo a2dissite 000-default.conf
Site 000-default disabled.
To activate the new configuration, you need to run:
    systemctl reload apache2
osboxes@khadimul-islam:~$ sudo a2ensite khadimul-islam.me.conf
Enabling site khadimul-islam.me.
To activate the new configuration, you need to run:
    systemctl reload apache2
osboxes@khadimul-islam:~$ sudo apache2 configtest
Usage: apache2 [-D name] [-d directory] [-f file]
               [-C "directive"] [-c "directive"]
               [-k start|restart|graceful|graceful-stop|stop]
               [-v] [-V] [-h] [-l] [-L] [-t] [-T] [-S] [-X]

Options:
  -D name          : define a name for use in <IfDefine name> directives
  -d directory     : specify an alternate initial ServerRoot
  -f file          : specify an alternate ServerConfigFile
  -C "directive"   : process directive before reading config files
  -c "directive"   : process directive after reading config files
  -e level         : show startup errors of level (see LogLevel)
  -E file          : log startup errors to file
  -v              : show version number
  -V              : show compile settings
  -h              : list available command line options (this page)
  -l              : list compiled in modules
  -L              : list available configuration directives
  -t -D DUMP_VHOSTS : show parsed vhost settings
  -t -D DUMP_RUN_CFG : show parsed run settings
  -S              : a synonym for -t -D DUMP_VHOSTS -D DUMP_RUN_CFG
  -t -D DUMP_MODULES : show all loaded modules
  -M              : a synonym for -t -D DUMP_MODULES
  -t -D DUMP_INCLUDES : show all included configuration files
  -t              : run syntax check for config files
  -T              : start without DocumentRoot(s) check
  -X              : debug mode (only one worker, do not detach)
osboxes@khadimul-islam:~$ sudo systemctl restart apache2
osboxes@khadimul-islam:~$
```

```

osboxes@khadimul-islam:~$ sudo nano /etc/apache2/sites-available/khadimul-islam.me.conf
osboxes@khadimul-islam:~$ sudo a2dissite 000-default.conf
Site 000-default disabled.
To activate the new configuration, you need to run:
    systemctl reload apache2
osboxes@khadimul-islam:~$ sudo a2ensite khadimul-islam.me.conf
Enabling site khadimul-islam.me.
To activate the new configuration, you need to run:
    systemctl reload apache2
osboxes@khadimul-islam:~$ sudo apache2 configtest
Usage: apache2 [-D name] [-d directory] [-f file]
               [-C "directive"] [-c "directive"]
               [-k start|restart|graceful|graceful-stop|stop]
               [-v] [-V] [-h] [-l] [-L] [-t] [-T] [-S] [-X]

Options:
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  -t -D DUMP_VHOSTS : show parsed vhost settings
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  -t -D DUMP_MODULES : show all loaded modules
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  -t -D DUMP_INCLUDES : show all included configuration files
  -t              : run syntax check for config files
  -T              : start without DocumentRoot(s) check
  -X              : debug mode (only one worker, do not detach)
osboxes@khadimul-islam:~$ sudo systemctl restart apache2
osboxes@khadimul-islam:~$ sudo apache2 -t
[Thu Nov 11 00:28:22.744645 2025] [core:warn] [pid 1848:tid 1848] AH00111: Config variable ${APACHE_RUN_DIR} is not defined
apache2: Syntax error on line 80 of /etc/apache2/apache2.conf: DefaultRuntimeDir must be a valid directory, absolute or relative to ServerRoot
osboxes@khadimul-islam:~$ sudo apache2ctl -t
Syntax OK
osboxes@khadimul-islam:~$ sudo bash -c 'grep -q "^export APACHE_RUN_DIR=" /etc/apache2/envvars || echo "export APACHE_RUN_DIR=/var/run/apache2" >> /etc/apache2/envvars'
bash: -c: line 1: unexpected EOF while looking for matching `"'
osboxes@khadimul-islam:~$ sudo bash -c 'grep -q "^export APACHE_RUN_DIR=" /etc/apache2/envvars || echo "export APACHE_RUN_DIR=/var/run/apache2" >> /etc/apache2/envvars'
osboxes@khadimul-islam:~$ sudo bash -c 'grep -q "^export APACHE_LOCK_DIR=" /etc/apache2/envvars || echo "export APACHE_LOCK_DIR=/var/lock/apache2" >> /etc/apache2/envvars'
osboxes@khadimul-islam:~$ sudo bash -c 'grep -q "^export APACHE_LOG_DIR=" /etc/apache2/envvars || echo "export APACHE_LOG_DIR=/var/log/apache2" >> /etc/apache2/envvars'
osboxes@khadimul-islam:~$ sudo bash -c 'grep -q "^export APACHE_PID_DIR=" /etc/apache2/envvars || echo "export APACHE_PID_FILE=${APACHE_RUN_DIR}/apache2.pid" >> /etc/apache2/envvars'
osboxes@khadimul-islam:~$ sudo mkdir -p /var/run/apache2
mkdir: invalid option -- '/'
Try 'mkdir --help' for more information.
osboxes@khadimul-islam:~$ sudo mkdir -p /var/run/apache2
osboxes@khadimul-islam:~$ sudo systemctl cat apache2 | grep -i EnvironmentFile || true
osboxes@khadimul-islam:~$ sudo systemctl daemon-reload
osboxes@khadimul-islam:~$ sudo systemctl restart apache2
osboxes@khadimul-islam:~$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-11-11 00:39:26 EST; 18s ago
     Invocation: 03e7846c276f421abf928e4153084ead
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 2038 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
      Main PID: 2042 (apache2)
         Tasks: 6 (limit: 1065)
        Memory: 10.8M (peak: 11M)
           CPU: 63ms
      CGroup: /system.slice/apache2.service
              └─2042 /usr/sbin/apache2 -k start
                └─2044 /usr/sbin/apache2 -k start
                  └─2045 /usr/sbin/apache2 -k start
                    └─2046 /usr/sbin/apache2 -k start
                      └─2047 /usr/sbin/apache2 -k start
                        └─2048 /usr/sbin/apache2 -k start

Nov 11 00:39:26 khadimul-islam.me systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 11 00:39:26 khadimul-islam.me systemd[1]: Started apache2.service - The Apache HTTP Server.
osboxes@khadimul-islam:~$ sudo apache2ctl -t
Syntax OK

```

```

osboxes@khadimul-islam:~$ curl -I http://khadimul-islam.me/phpmyadmin
HTTP/1.1 301 Moved Permanently
Date: Tue, 11 Nov 2025 05:48:02 GMT
Server: Apache/2.4.62 (Ubuntu)
Location: http://khadimul-islam.me/phpmyadmin/
Content-Type: text/html; charset=iso-8859-1

osboxes@khadimul-islam:~$ sudo ss -tulpen | egrep '(:80|:443|:3306)'
tcp LISTEN 0      70      127.0.0.1:33060      0.0.0.0:*      users:(("mysqld",pid=869,fd=21))
                                uid:115 ino:9721 sk:7 cgroup:/system.slice/mysql.service <->
tcp LISTEN 0      151     127.0.0.1:3306      0.0.0.0:*      users:(("mysqld",pid=869,fd=33))
                                uid:115 ino:9829 sk:d cgroup:/system.slice/mysql.service <->
tcp LISTEN 0      511     *:80                *:.*           users:(("apache2",pid=6303,fd=4),("apache2",pid=6302,fd=4),("apache2",pid=6299,fd=4),("apache2",pid=6242,fd=4)) ino:27250 sk:10 cgroup:/system.slice/apache2.service v6only:0 <->
osboxes@khadimul-islam:~$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-11-11 00:45:03 EST; 6min ago
  Invocation: 68209d5027664fe6812e8043cca91514
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 6239 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Process: 6294 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
  Main PID: 6242 (apache2)
     Tasks: 6 (limit: 1065)
  Memory: 17.2M (peak: 28.7M swap: 1.3M swap peak: 1.3M)
     CPU: 501ms
    CGroup: /system.slice/apache2.service
            └─6242 /usr/sbin/apache2 -k start
              └─6299 /usr/sbin/apache2 -k start
                └─6300 /usr/sbin/apache2 -k start
                  └─6301 /usr/sbin/apache2 -k start
                    └─6302 /usr/sbin/apache2 -k start
                      └─6303 /usr/sbin/apache2 -k start

Nov 11 00:45:03 khadimul-islam.me systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 11 00:45:03 khadimul-islam.me systemd[1]: Started apache2.service - The Apache HTTP Server.
Nov 11 00:45:57 khadimul-islam.me systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Nov 11 00:45:57 khadimul-islam.me systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
osboxes@khadimul-islam:~$

```

```

adding: etc/apache2/mods-available/session_cookie.load (deflated 27%)
adding: etc/apache2/mods-available/headers.load (deflated 21%)
adding: etc/apache2/mods-available/proxy_http.load (deflated 24%)
adding: etc/apache2/mods-available/proxy_uwsgi.load (deflated 24%)
adding: etc/apache2/mods-available/auth_digest.load (deflated 22%)
adding: etc/apache2/mods-available/log_debug.load (deflated 23%)
adding: etc/apache2/mods-available/status.load (deflated 20%)
adding: etc/apache2/mods-available/info.conf (deflated 33%)
adding: etc/apache2/mods-available/allowmethods.load (deflated 25%)
adding: etc/apache2/mods-available/unique_id.load (deflated 23%)
adding: etc/apache2/mods-available/log_forensic.load (deflated 25%)
adding: etc/apache2/mods-available/authz_owner.load (deflated 24%)
adding: etc/apache2/mods-available/suexec.load (deflated 20%)
adding: etc/apache2/mods-available/proxy_ftp.conf (deflated 5%)
adding: etc/apache2/mods-available/authz_user.load (deflated 23%)
adding: etc/apache2/mods-available/authz_core.load (deflated 24%)
adding: etc/apache2/mods-available/proxy_ajp.load (deflated 23%)
adding: etc/apache2/mods-available/actions.conf (deflated 37%)
adding: etc/apache2/mods-available/mpm_worker.load (deflated 22%)
adding: etc/apache2/mods-available/proxy_connect.load (deflated 25%)
adding: etc/apache2/mods-available/userdir.conf (deflated 23%)
adding: etc/apache2/mods-available/lbmethod_heartbeat.load (deflated 23%)
adding: etc/apache2/mods-available/ssl.conf (deflated 52%)
adding: etc/apache2/mods-available/file_cache.load (deflated 25%)
adding: etc/apache2/mods-available/mpm_worker.conf (deflated 60%)
adding: etc/apache2/mods-available/proxy_ftp.load (deflated 23%)
adding: etc/apache2/mods-available/imagemap.load (deflated 22%)
adding: etc/apache2/mods-available/heartbeat.load (deflated 27%)
adding: etc/apache2/mods-available/deflate.load (deflated 17%)
adding: etc/apache2/mods-available/cache.load (deflated 23%)
adding: etc/apache2/mods-available/md.load (deflated 16%)
adding: etc/apache2/mods-enabled/ (stored 0%)
adding: etc/apache2/mods-enabled/mime.conf (deflated 62%)
adding: etc/apache2/mods-enabled/status.conf (deflated 42%)
adding: etc/apache2/mods-enabled/env.load (deflated 17%)
adding: etc/apache2/mods-enabled/mime.load (deflated 18%)
adding: etc/apache2/mods-enabled/deflate.conf (deflated 62%)
adding: etc/apache2/mods-enabled/dir.conf (deflated 43%)
adding: etc/apache2/mods-enabled/php8.3.conf (deflated 50%)
adding: etc/apache2/mods-enabled/mpm_prefork.load (deflated 23%)
adding: etc/apache2/mods-enabled/alias.conf (deflated 46%)
adding: etc/apache2/mods-enabled/authz_host.load (deflated 23%)
adding: etc/apache2/mods-enabled/php8.3.load (deflated 13%)
adding: etc/apache2/mods-enabled/authn_core.load (deflated 24%)
adding: etc/apache2/mods-enabled/negotiation.conf (deflated 42%)
adding: etc/apache2/mods-enabled/alias.load (deflated 19%)
adding: etc/apache2/mods-enabled/setenvif.load (deflated 22%)
adding: etc/apache2/mods-enabled/filter.load (deflated 20%)
adding: etc/apache2/mods-enabled/reqtimeout.load (deflated 24%)
adding: etc/apache2/mods-enabled/access_compat.load (deflated 21%)
adding: etc/apache2/mods-enabled/setenvif.conf (deflated 60%)
adding: etc/apache2/mods-enabled/negotiation.load (deflated 24%)
adding: etc/apache2/mods-enabled/dir.load (deflated 17%)
adding: etc/apache2/mods-enabled/reqtimeout.conf (deflated 51%)
adding: etc/apache2/mods-enabled/authn_file.load (deflated 24%)
adding: etc/apache2/mods-enabled/mpm_prefork.conf (deflated 63%)
adding: etc/apache2/mods-enabled/auth_basic.load (deflated 21%)
adding: etc/apache2/mods-enabled/autodisk.conf (deflated 63%)
adding: etc/apache2/mods-enabled/autodisk.load (deflated 23%)
adding: etc/apache2/mods-enabled/status.load (deflated 20%)
adding: etc/apache2/mods-enabled/authz_user.load (deflated 23%)
adding: etc/apache2/mods-enabled/authz_core.load (deflated 24%)
adding: etc/apache2/mods-enabled/deflate.load (deflated 17%)
adding: etc/apache2/sites-available/ (stored 0%)
adding: etc/apache2/sites-available/default-ssl.conf (deflated 59%)
adding: etc/apache2/sites-available/000-default.conf (deflated 46%)
adding: etc/apache2/sites-available/khadimul-islam.me.conf (deflated 38%)
adding: etc/apache2/ports.conf (deflated 35%)
adding: etc/apache2/sites-enabled/ (stored 0%)
adding: etc/apache2/sites-enabled/khadimul-islam.me.conf (deflated 38%)
adding: etc/apache2/conf-available/ (stored 0%)
adding: etc/apache2/conf-available/security.conf (deflated 51%)
adding: etc/apache2/conf-available/javascript-common.conf (deflated 27%)
adding: etc/apache2/conf-available/other-vhosts-access-log.conf (deflated 13%)
adding: etc/apache2/conf-available/charset.conf (deflated 33%)
adding: etc/apache2/conf-available/serve-cgi-bin.conf (deflated 47%)
adding: etc/apache2/conf-available/phpmyadmin.conf (deflated 66%)
adding: etc/apache2/conf-available/localized-error-pages.conf (deflated 59%)
osboxes@khadimul-islam:~$ ls -lh ~apache2-config.zip
ls: cannot access '~apache2-config.zip': No such file or directory
osboxes@khadimul-islam:~$ ls -lh ~/apache2-config.zip
-rw-r--r-- 1 root root 92K Nov 11 00:55 /home/osboxes/apache2-config.zip

```





